Bruno Luçardo Olivaes

Independent Researcher · Computational Physics & Quantum Metrology

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Research Summary

Independent computational physicist and research engineer investigating the foundations of scale-invariance in physical systems, observational cosmology, quantum error mitigation on NISQ hardware, neural simulation-based inference at particle colliders, and post-quantum network security. Author of **The Matrioska Protocol**, a pre-registered statistical architecture enforcing 5-gate anti-hallucination standards across heterogeneous experimental datasets.

Unified Master Monograph

The Matrioska Protocol: From Primordial Discrete Scale Invariance to Post-Quantum Agility
Monograph Synthesis (34 pages), November 2026. CDD: 500 → 530.1.
Permanent CERN / Zenodo Archive: doi:10.5281/zenodo.24124702.

Scientific Publications & Preprints

1. **Breaking the Zeptouniverse Sensitivity Barrier: Neural Simulation-Based Inference for Certified Discrete Scale Invariance Searches at the LHC**
   B. L. Olivaes (2026). CDD: 006.3 / 539.7.
   Physical Review D (in submission). GitHub: bruno-olivaes/lhc-nsbi-dsi. Release: v1.0.0.

2. **A Certified Search for Discrete Scale Invariance in LHC Dimuon Mass Spectra**
   B. L. Olivaes (2026). CDD: 539.7 / 006.3.
   CMS Open Data ($\sqrt{s} = 7\,TeV$, $N = 62,214$). GitHub: bruno-olivaes/lhc-dsi-search. Release: v1.0.0.

3. **Constraints on Primordial Discrete Scale Invariance from the Planck 2018 Temperature Power Spectrum**
   B. L. Olivaes (2026). CDD: 523.1 / 530.1.
   Monthly Notices of the Royal Astronomical Society (in submission). Pre-registered (OSF).
   Permanent Zenodo Archive: doi:10.5281/zenodo.24121677. GitHub: bruno-olivaes/dsi-cosmo-search.

4. **Statistical Pitfalls in Casimir Force Spectroscopy: Eliminating False-Positive Periodic Signals in Sub-Micron AFM Metrology**
   B. L. Olivaes (2026). CDD: 530.4 / 620.5.
   Review of Scientific Instruments (in submission).
   Permanent Zenodo Archive: doi:10.5281/zenodo.24121687. GitHub: bruno-olivaes/casimir-afm-spectroscopy.

5. **When Harmonic ZNE Fails: A Comparative Benchmark of Error Mitigation on Superconducting NISQ Processors**
   B. L. Olivaes (2026). CDD: 006.3 / 530.1.
   Quantum Science and Technology (in submission).
   Permanent Zenodo Archive: doi:10.5281/zenodo.24121693. GitHub: bruno-olivaes/qiskit-lombscargle-zne.

6. **Forecasting Log-Oscillatory Dark Energy with ESA Euclid: Breaking the Degeneracy**
   B. L. Olivaes (2026). CDD: 523.1. Joint Pantheon+ (1580×1580 cov), DESI BAO, and Planck CMB.
   Permanent Zenodo Archive: doi:10.5281/zenodo.24121691. GitHub: bruno-olivaes/oscillatory-dark-energy-euclid.

7. **The Network Cost of Post-Quantum Cryptography: TCP Fragmentation and Handshake Latency in TLS 1.3**
   B. L. Olivaes (2026). CDD: 005.8 / 006.3. Benchmarking NIST FIPS 203 (ML-KEM-768) wire overhead.
   Permanent Zenodo Archive: doi:10.5281/zenodo.24121695. GitHub: bruno-olivaes/pqc-crypto-agility-benchmark.

Technical Competencies & Software Architecture

- **Scientific Programming & Data Analysis:** Python 3.12+, NumPy, SciPy (optimize, signal), Astropy (LombScargle), Matplotlib, Pandas, HDF5/h5py, PyMuPDF vector extraction.
- **Machine Learning & Statistical Inference:** PyTorch 2.x, Neural Simulation-Based Inference (NSBI), continuous neural likelihood estimation, Gauss-Legendre quadrature, profile likelihood ($q(\varepsilon)$), MCMC sampling (`emcee`).
- **Quantum Computing:** Qiskit 2.x, Qiskit Runtime Primitives (`EstimatorV2`), Zero-Noise Extrapolation (ZNE), Pauli Twirling / Randomized Compiling, fake heavy-hex backends.
- **Post-Quantum Cryptography & Networking:** NIST FIPS 203 (ML-KEM/Kyber), liboqs, TLS 1.3 packet inspection, Wireshark, TCP fragmentation analysis.
- **Rigorous Metrology & Open Science:** Pre-registration workflows (OSF), 5-Gate Integrity Protocol, look-elsewhere correction via phase-scrambled bootstrap ($B = 1000$), Dewey Decimal Classification (DDC) taxonomies, Zenodo/CERN integration, Git/GitHub, LaTeX.